

TECHNICAL PROPOSAL · COMPLETE PLATFORM

# Backend + Cloud: a platform that's **real, safe** **and always on.**

Strolla Health's Flutter app is finished and polished — but every social feature runs on placeholder data and nothing survives outside one phone. This proposal covers the two halves of going live: a **Python backend** that powers every existing screen, and **professional DevOps on AWS** so the service is secure, monitored, backed up and ready for thousands of UK users from day one.

**2**

workstreams: Backend  
(Python) + DevOps (AWS)

**UK**

hosted in AWS London region  
— UK data residency, GDPR-  
aligned

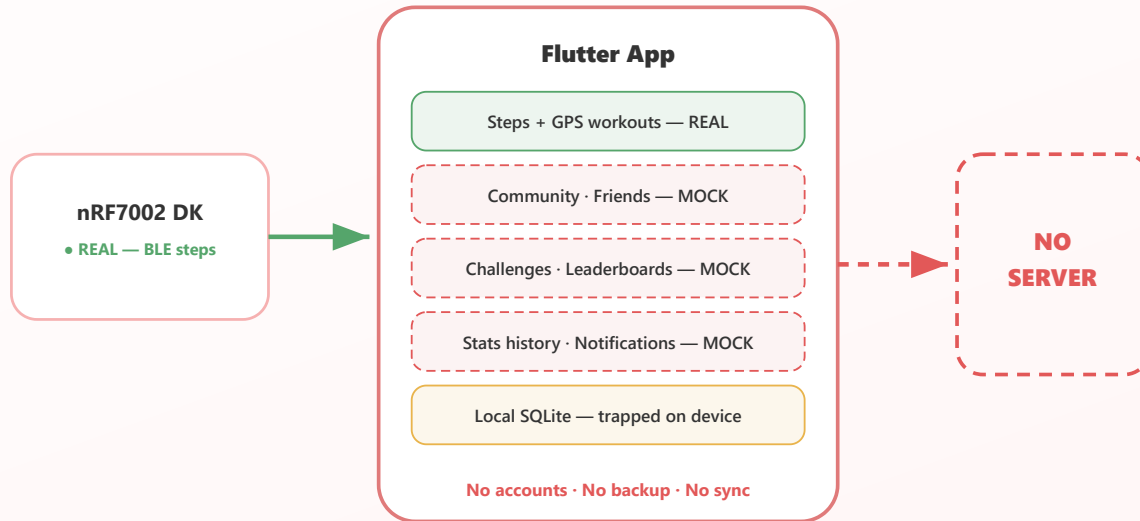
**99.9%**

availability target with multi-  
zone failover

**0**

downtime deployments —  
users never notice a release

## A finished app with **nothing** behind it



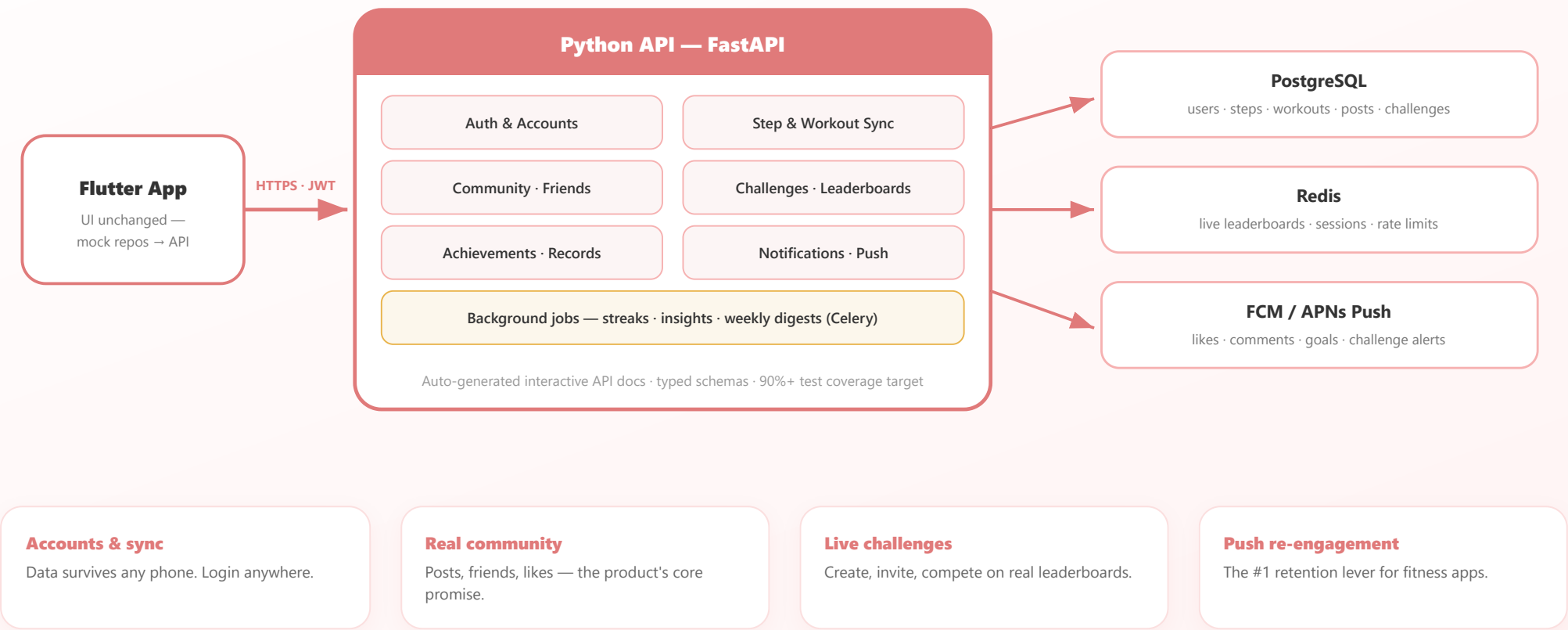
### What's at stake

- ✗ Delete the app → **all data gone forever**
- ✗ Every user sees the same fake friends & posts
- ✗ No way to run challenges or campaigns
- ✗ No visibility: crashes & usage are invisible to us

### And even with a backend...

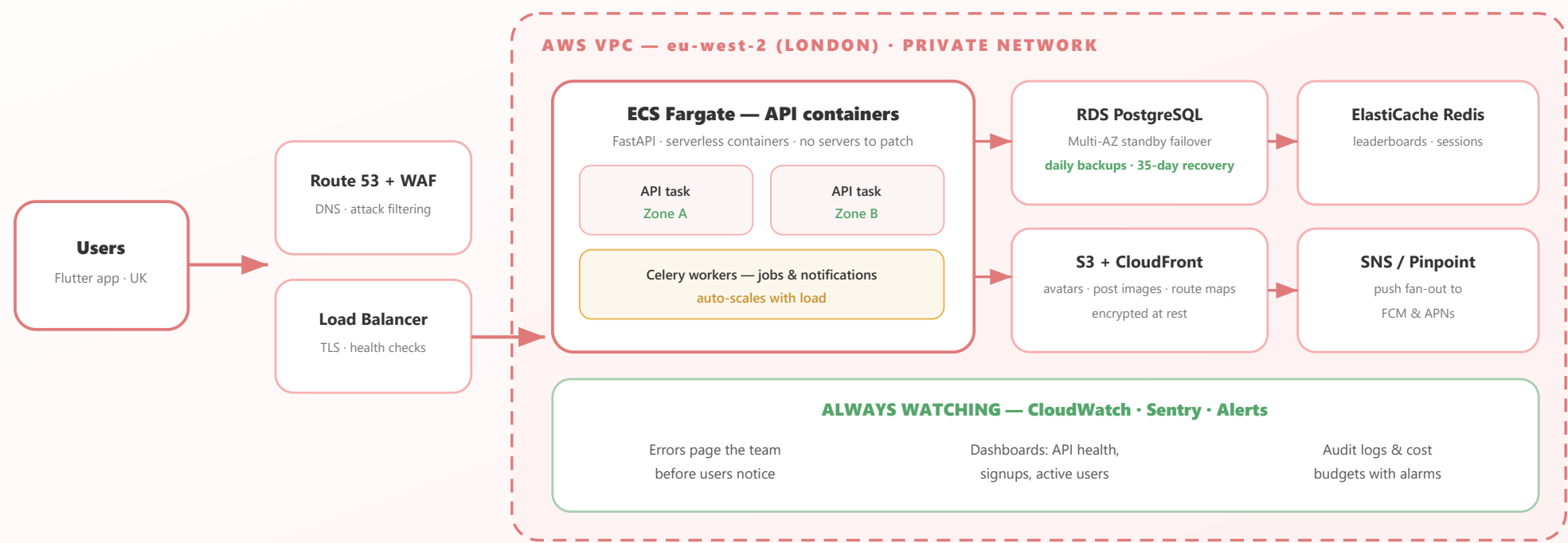
a server that's deployed by hand onto a single machine **fails alone at 2 a.m.**, loses data without backups, and breaks with every release. That's why this proposal pairs the backend with a proper cloud foundation — **the two halves only make sense together.**

# A Python platform that powers every screen already built



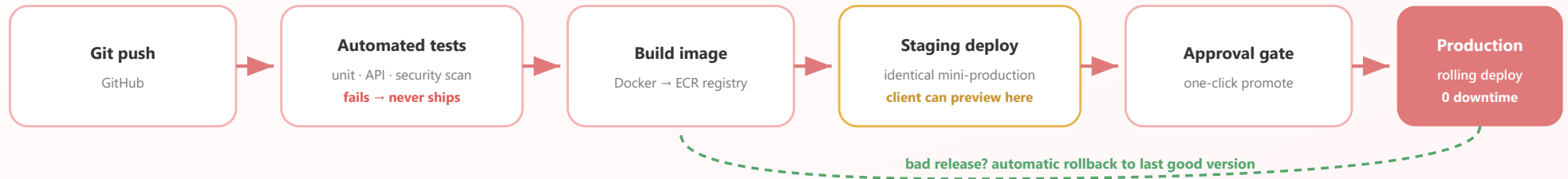
# Production architecture on AWS — London region

Managed, multi-zone, auto-scaling. UK user data stays in the UK (eu-west-2) — a genuine trust and GDPR advantage for a women's health brand.



# From code to production in minutes, with zero downtime

Every change is tested, reviewed and deployed automatically. Releases stop being events — they become routine.



## Three environments

| ENV        | PURPOSE                                    |
|------------|--------------------------------------------|
| Dev        | Developers iterate freely, breaks allowed  |
| Staging    | Exact production clone — client demos & QA |
| Production | Real users · locked down · monitored 24/7  |

All three defined as code (Terraform) — rebuildable from scratch in under an hour.

## Security & GDPR

- ✓ Data encrypted in transit (TLS) and at rest
- ✓ UK data residency — AWS London region
- ✓ Right-to-erasure & data export endpoints
- ✓ Secrets in AWS Secrets Manager, never in code
- ✓ WAF blocks common attacks at the edge

## Resilience

- ✓ Two availability zones — one can fail entirely
- ✓ Database failover in under 60 seconds
- ✓ Daily backups, 35-day point-in-time restore
- ✓ Auto-scaling absorbs press-day traffic spikes
- ✓ Alerts reach the team before users tweet

## Backend alone vs. **backend + DevOps**

The backend creates the product. DevOps keeps the promise the product makes — every day, at any scale.

### Backend on a "simple server"

- ✗ One machine — when it fails, **the app is down** until someone wakes up
- ✗ Manual deploys — each release is risky, late-night work
- ✗ Backups only if someone remembers
- ✗ No staging — testing happens on real users
- ✗ A traffic spike (press, influencer post) takes the service offline
- ✗ No monitoring — we learn about outages from app-store reviews

### With the AWS DevOps workstream

- ✓ Multi-zone — a whole data centre can fail, users never notice
- ✓ Push-button releases with automatic rollback
- ✓ Daily automated backups, 35-day recovery window
- ✓ Staging clone for demos & QA before anything ships
- ✓ Auto-scaling — launch-day spikes are absorbed, then costs scale back down
- ✓ 24/7 monitoring with alerts — we fix issues before users see them

**< 60s**

database failover if a zone goes down

**~10 min**

from approved code to live in production

**35 days**

point-in-time data recovery window

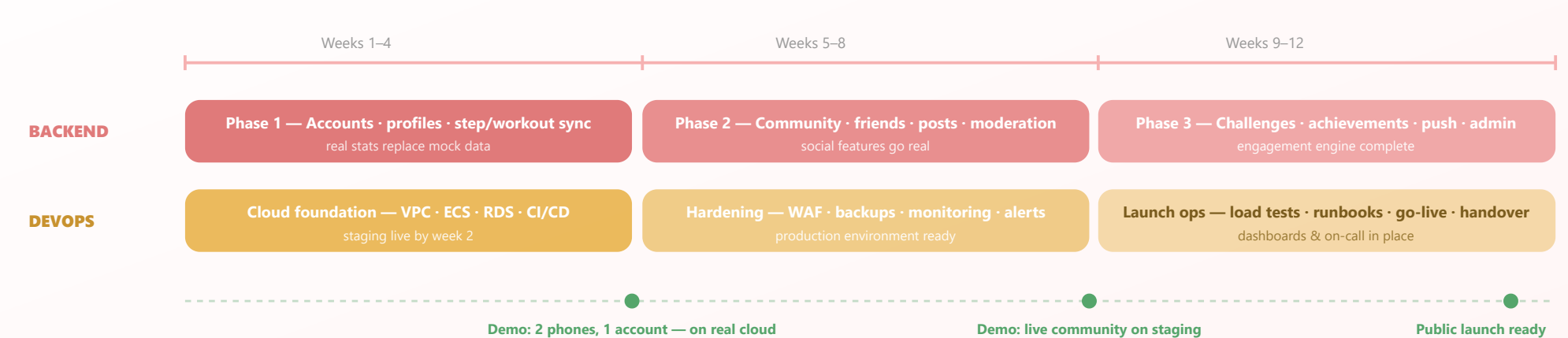
**Pay-as-you-grow**

Fargate + managed services scale with real usage — small at launch, no re-platforming later

## DELIVERY PLAN

# One roadmap, two tracks — **live infrastructure from week 2**

DevOps runs in parallel with backend development, so every phase is demoed on real cloud infrastructure, not a laptop.



### What you receive at handover

- ✓ Full backend source code & API documentation
- ✓ Infrastructure as code (Terraform) — the whole platform is reproducible
- ✓ CI/CD pipelines with automated tests
- ✓ Monitoring dashboards & alerting
- ✓ Admin panel for users, content & metrics
- ✓ Runbooks + optional ongoing support retainer

### The bottom line

The frontend investment is already made. Backend + AWS DevOps is what turns it into a **trustworthy, always-on product** — real community, protected data, and infrastructure that scales from first user to first hundred thousand without re-building.